Technology
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SUSTAINABLE SUPPLY CHAIN MANAGEMENT
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RhineSound

A qualified backup for port disruption

When the China supply route fails, how do output, unit cost, CO_2e and shortage downtime change, and which response helps?

01 JBL Flip 6: battery supply and the five-station line

CHINA
PKCELL*

DE backup*
VOLTRONIC

Yantian / sea freight
67 ± 11 workdays

10 ± 2 workdays

Cologne stock
1,106 packs safety stock

01 **Housing & drivers** 3 min → 02 **Electronics** 4 min → 03 **Battery installation** 3 min → 04 **Firmware & test** 6 min → 05 **Inspect & pack** 2 min

52 weeks · 40 h/week · Demand: 300 units/week · 90% availability · Setup: 10 min/100 units

*Real supplier candidates. Pack compatibility, purchasing terms and Cologne assembly are model assumptions.

02 Disruption and countermeasure

3 weeks

Export-port access stops in weeks 21–23.
Cargo already past the port keeps moving.

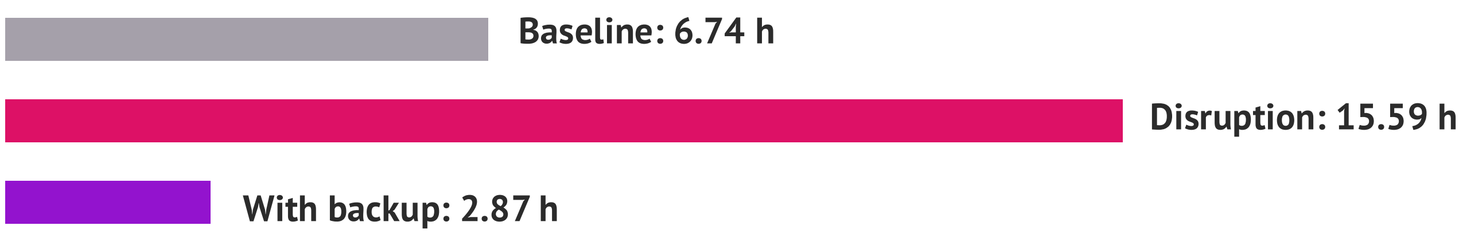
900 backup packs

Order from the DE supplier after one workday.
Its own lead time applies. Qualification: €300/year.

Real case: COVID restrictions at Yantian in 2021. Full operations resumed on 24 June [3].
The exact three-week full block is hypothetical. The backup requires prior qualification [1].

03 Three scenarios: less downtime, unchanged annual output

Shortage downtime at the battery station (hours/year)



Ten seeds (1–10) per scenario. Downtime min–max: baseline 0–28.73 h, disruption 0–88.53 h, with backup 0–28.69 h.

−82 %
less shortage downtime
+€0.28/unit vs disruption.
Keep the DE backup qualified.

Animation
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Code
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299.96
units/week in all three scenarios
Mean = min = max; losses recover within the year [2].

Baseline	Disruption	With backup
Cost: €53.03/unit	**Cost: €53.03/unit**	**Cost: €53.31/unit**
Min–max 53.00–53.05 €	Min–max 53.00–53.05 €	Min–max 53.27–53.33 €
1.4327 kg CO_2e/unit	**1.4327 kg CO_2e/unit**	**1.4308 kg CO_2e/unit**
Min–max 1.4326–1.4327 kg	Min–max 1.4326–1.4327 kg	Min–max 1.4307–1.4308 kg

Course concepts explain the results

Bottleneck, variability and Little's Law

S4 capacity incl. setups: 354 units/week > demand 300.
Random times create queues. Spare capacity allows recovery.
Baseline seed 1: L = 13.48 ≈ 7.499 units/h × 1.80 h.

Lead time, safety stock and sourcing

Safety stock buffers long, variable China lead times.
Qualified DE supply cuts mean starvation by 12.72 h.
Whitney et al. [1]: alternatives need suitable capability.
Ivanov [2]: assess time paths as well as network links.

Annual cost: materials + 52 × weekly fixed costs + freight + 20% of mean stock value + €100/tCO_2e + backup qualification.
Baseline €53.03/unit < €117.64 net list price (illustrative retail comparison). Cost per unit = annual cost / annual output.
Partial CO_2e footprint: battery production, battery transport and factory electricity. DE supply cuts transport emissions.

[1] Whitney, D. E., Luo, J., & Heller, D. A. (2014). The benefits and constraints of temporary sourcing diversification in supply chain disruption and recovery. *Journal of Purchasing and Supply Management, 20*(4), 238–250. https://doi.org/10.1016/j.pursup.2014.06.001

[2] Ivanov, D. (2025). What network and performance indicators can tell us about supply chain and sourcing resilience (and what they cannot). *Journal of Purchasing and Supply Management, 31*(3), 101014. https://doi.org/10.1016/j.pursup.2025.101014

[3] YANTIAN (2021, June 23). YANTIAN Resumes Full Operations. yict.com.cn/article/detail/8948.html. Product photo: JBL. Icons: Lucide (ISC).